DP Computer Science: A 1.1.4 Primary Memory

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1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

1. Explain the purpose of **RAM, ROM, cache, and registers**, giving a reason for the way each is built, including:
    - Why **RAM is volatile**
    - Why **ROM is read-only**
    - Why **cache exists at all**
2. Explain the **order** in which the CPU searches primary memory, from registers down to RAM, and why that order keeps the CPU waiting as little as possible
3. Explain what a **cache hit** and a **cache miss** are, and why a **higher hit rate** improves performance

> **Command Term Note:** Each intention lands at **Explain**—a detailed account including reasons or causes. Naming what each type stores is only description and sits below the statement.

Key Vocabulary (Tier 3)

Term	Definition
Volatile	Loses its contents when the power is switched off
Non-volatile	Keeps its contents without power
Register	A tiny, very fast storage location inside the CPU, holding what it is working on this instant
Memory hierarchy	The ordering of memory types by speed, from registers down to RAM, trading capacity for speed at each level
Cache hit	The CPU finds the data it needs in cache and proceeds almost immediately

Term	Definition
Cache miss	The data is not in cache, so the CPU fetches it from RAM, which takes much longer

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Pedagogical Focus

- **Thinking Skills (ATL):** The primary skill students consciously practise is **transfer**—taking the registers they already named in A1.1.1 and the concrete cases taught here, and abstracting them into one general model of the memory hierarchy.
- **Conceptual Understanding (ATT):** Teach this lesson **explicitly and in order**, not by discovery. Start from a concrete example students already hold (registers), teach the other three types the same way, then abstract upward to the single hierarchy model before students apply it.

Name the skill to students: "*You are now connecting four separate facts into one ordered model.*"

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2. Lesson Sequence (One-Hour Block)

Part 1: Do Now (Retrieval) — ~10%

Activity: Three questions recalling prior learning:

Question	Source
1. Name the register that holds the address of the next instruction to be fetched	[A1.1.1]
2. During the fetch stage, which register puts an address onto the address bus, and which register receives the instruction that comes back?	[A1.1.5, A1.1.1]
3. State the three stages of the machine instruction cycle, in order	[A1.1.5]

> **Note:** Registers reappear today as the fastest memory, so they are the retrieval bridge.

Part 2: Main Sequence — ~60%

Phase A: Concrete Example with Explicit Teaching (~6 mins)

Begin with Registers (the concrete case students already know):

- Fastest memory
- Inside the CPU
- Holds exactly what it is working on now

Then teach RAM as concrete working memory:

- **Purpose:** Stores active data and programs currently in use
- **Reason for volatility:** Everything in RAM is a working copy; losing it at power-off is acceptable because permanent versions live in storage

> **Teaching Principle:** Keep the reason attached to each type, not just the label.

Phase B: Progressive Abstraction (~8 mins)

Add ROM:

Feature	Explanation
Purpose	Stores essential system startup instructions (BIOS/UEFI)
Reason for read-only	Startup instructions must survive power-off and must not be corrupted by running software
Reason for non-volatility	Instructions must be available instantly when the machine is switched on

Add Cache:

Feature	Explanation
Purpose	Small, faster staging area between CPU and RAM

Feature	Explanation
Reason cache exists	RAM, fast as it is, still keeps the CPU waiting; a small, faster staging area reduces this delay
Levels	L1 (smallest, fastest per core) → L2 (larger, slightly slower) → L3 (largest, shared across cores)

Now Generalise — Draw Out the Underlying Principle:

> *The faster the memory, the more it costs per byte and the less can sit near the CPU. This is why memory is ordered as a hierarchy.*

Phase C: Linking Task (~4 mins)

Task: Students draw the hierarchy from registers down to RAM:

```text
┌─────────────────────────────────────────────┐
│  REGISTERS     ← Fastest, smallest, most expensive │
├─────────────────────────────────────────────┤
│  L1 CACHE      ← Fast, small, per core           │
├─────────────────────────────────────────────┤
│  L2 CACHE      ← Medium speed, medium size       │
├─────────────────────────────────────────────┤
│  L3 CACHE      ← Slower, larger, shared          │
├─────────────────────────────────────────────┤
│  RAM           ← Slowest, largest, cheapest       │
└─────────────────────────────────────────────┘
```

On the same diagram, add the hit/miss layer:

- **Cache hit:** Data found in cache → proceed almost immediately
- **Cache miss:** Data not in cache → fetch from RAM (takes much longer)

Part 3: Deliberate Practice — ~20%

Activity A — "Cache Hit or Miss?" Scenarios:

For each scenario, decide **hit or miss** and explain the performance consequence in one sentence:

Scenario	Prediction	Explanation
Frequently used program instructions	**Hit**	Repeated access means data stays in cache
A large image loaded once	**Miss**	Data not previously accessed; must fetch from RAM
A repeated calculation	**Hit**	Once loaded, it stays in cache for reuse

Sentence Frame:

> "*This scenario would result in a [hit/miss] because [reason], meaning performance is [improved/reduced] since [explanation].*"

Activity B — Written Explain Answer:

> "*Explain why the CPU searches registers and cache before RAM.*"

Expected Reason:

> Each step down is larger but slower, so keeping the soonest-needed data in the fastest level makes the slow trips as rare as possible.

Part 4: Exit Check — ~10%

One Explain Question (2 marks):

> "*Explain why ROM rather than RAM is used to store the instructions a computer needs at startup.*"

Mark Scheme:

Marks	Criteria
[1]	RAM is volatile, so anything in it is lost at power-off, but startup instructions must be available the instant the machine is switched on

Marks	Criteria
[1]	ROM is non-volatile and read-only, so the instructions survive without power and cannot be overwritten or corrupted by running software

> **Teacher Note:** Read these the same day. A student who names ROM as non-volatile but gives no reason has stopped at **Describe** and needs the "because" prompt next lesson.

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3. Summary: Types of Primary Memory

Memory Type	Purpose	Key Characteristics
Registers	Hold data the CPU is actively using	Fastest, smallest, inside CPU
Cache (L1, L2, L3)	Speed up data access by staging between CPU and RAM	Small, fast, multiple levels
RAM	Store active data and programs	Volatile, larger, slower than cache
ROM	Hold essential system startup instructions	Non-volatile, read-only

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4. Differentiation & Extension

Support (Scaffolding That Can Be Removed)

Support Strategy	Description
Pre-filled Hierarchy Diagram	Provide with labels; student adds reasons
Sentence Starters	For the "Cache Hit or Miss?" scenarios
Partially Completed Table	Memory type and purpose filled in; reason left blank

Challenge (Deeper or Wider)

Challenge Strategy	Description
Cache Replacement Policies	Research LRU, FIFO, and other cache replacement policies
Virtual Memory	Explore the concept and how it extends the memory hierarchy
Real-World Analysis	Research and compare memory specifications of different devices (smartphones, laptops, servers)

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5. Engagement Activities (Supplementary)

If time allows, these can replace or supplement the main sequence:

Activity	Description
"Memory Metaphors" Brainstorm	Students brainstorm analogies for computer memory (library, filing cabinet, backpack)
"Memory Hierarchy Relay Race"	Kinesthetic activity: team members represent different memory levels and pass information along
"Explain in Simple Words"	Teams explain their assigned memory type as if to a non-specialist
"Cache Hit or Miss?" Scenario Analysis	Present scenarios and discuss implications for performance
"Memory Optimisation" Brainstorm	Discuss strategies for optimising memory usage and improving cache hit rates

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6. Learning Objectives (Summary)

Knowledge and Understanding

- Define and differentiate between RAM, ROM, cache (L1, L2, L3), and registers
- Explain the specific purposes and functions of each type of primary memory
- Describe how the CPU interacts with different memory types to optimise performance
- Explain the concepts of cache hit and cache miss and their relevance to performance

Skills

- Analyse the hierarchical structure of primary memory and its influence on access speed
- Evaluate trade-offs between speed, size, and cost for different memory types
- Apply understanding of cache hits and misses to analyse memory access patterns

ATL Skills

Skill	Description
Communication	Effectively communicate understanding through discussions and written explanations
Collaboration	Work collaboratively to analyse diagrams, solve problems, and complete activities
Critical Thinking	Analyse relationships between memory types; evaluate their importance in computer systems
Information Literacy	Use online resources and research to deepen understanding

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7. Resources

- **Presentation** (Slide Deck)
- **InThinking eBook** (A 1.1.4)
- **Quizlet** (A 1.1.4 vocabulary flashcards)
- **Worksheet** (Memory hierarchy diagram)

Author's Reflection

> *"A1.1.4 is one of only two Explain statements in this subtopic—reasons and causes are introduced. The objectives from the previous lesson drifted up into evaluating trade-offs and optimising access; that is AO3, not what this statement asks, so I have cut it. This lesson is not inquiry-based, like the next lesson A1.1.5 which runs on the simulator. The memory hierarchy is a model students meet for the first time here, and inquiry would ask them to discover facts they cannot reasonably infer.*

> *So the four types of memory (RAM, ROM, registers, and cache) come first, each with its reason attached, then the single ordered model, then application. The linking task is the point: four separate facts becoming one hierarchy they can reason from."*